RAG Comparison: Dense vs Sparse vs Hybrid Retrieval Analysis

NLP Retrieval-Augmented Generation Benchmark
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Abstract

Retrieval-Augmented Generation (RAG) systems depend critically on whether the retriever can place answer-bearing evidence into the generator context. This report presents an empirical comparison of dense semantic retrieval, sparse lexical retrieval, and reciprocal-rank hybrid retrieval under a shared extractive question-answering generator, evaluated on 300 queries per dataset across two complementary benchmarks: SQuAD and TriviaQA.

On SQuAD, where questions are written with the source passage visible and therefore have high lexical overlap, sparse BM25 and hybrid RRF retrieval achieve the strongest F1 scores (0.701 and 0.695), while dense retrieval lags at 0.641. On TriviaQA, where questions are trivia-style and paraphrased, dense and hybrid retrieval lead (F1 0.458 and 0.465), while sparse drops to 0.429. The generalisation gap, measured as TriviaQA F1 minus SQuAD F1, is smallest for dense retrieval (-0.183) and largest for sparse retrieval (-0.272), making dense the most robust individual retriever across dataset types. Hybrid retrieval is the best overall practical choice, achieving the highest mean F1 across both datasets (0.580) with only small latency differences at this corpus scale. Failure-mode analysis shows that TriviaQA is limited mainly by the answer generator: 40–43% of queries retrieve answer-bearing evidence but still fail generation.

1 Introduction

Retrieval-Augmented Generation decomposes open-domain question answering into two linked decisions: first, retrieve a small set of passages likely to contain the answer; second, condition a reader or generator on those passages. This architecture makes retrieval quality visible. If the answer is absent from the retrieved context, even a strong answer model has little chance of producing a faithful response. Conversely, if the answer is present but the model still fails, the bottleneck lies in extraction, context formatting, or reader calibration.

The project analyzed here compares three retrieval strategies under the same answer generator: dense retrieval using SentenceTransformers and FAISS, sparse retrieval using BM25 Okapi, and hybrid retrieval via Reciprocal Rank Fusion (RRF). The central question is not merely which pipeline scores highest, but which pipeline offers the best practical balance between answer quality, retrieval robustness, and latency.

A key design decision is the use of two complementary datasets. SQuAD questions are written while looking at the source passage, creating high lexical overlap that theoretically favours BM25. TriviaQA questions are trivia-style and independently composed, requiring semantic matching that should favour dense retrieval. Running both datasets together tests whether findings generalise beyond a single favourable condition. The full experiment uses 300 queries per dataset, giving bootstrap 95% confidence intervals tight enough to draw meaningful comparative conclusions.

2 Background and Motivation

2.1 Dense Retrieval

Dense retrieval maps queries and passages into a shared vector space. In this project, passages are embedded using `sentence-transformers/all-MiniLM-L6-v2`. Vectors are L2-normalised and indexed using FAISS `IndexFlatIP`, so inner-product search corresponds to cosine similarity. Dense retrieval is attractive because it can match semantically related expressions even when they share few words, such as connecting "place of birth" with "was born in".

Its weakness is dependence on the embedding model's learned representations. It can miss exact rare names, numbers, or domain-specific terminology that are not well represented in the model. It also introduces query encoding cost compared with BM25.

2.2 Sparse Retrieval

Sparse retrieval uses BM25 Okapi from `rank-bm25`. BM25 ranks chunks by lexical overlap, term frequency, inverse document frequency, and document-length normalization. It is a strong baseline for SQuAD-style QA because SQuAD questions are written with the source passage visible, creating high word overlap between question and evidence.

BM25 is interpretable, fast, and requires no training. Its main limitation is vocabulary mismatch: when questions paraphrase the evidence, use synonyms, or refer to entities indirectly, lexical overlap alone is insufficient.

2.3 Hybrid Retrieval

The hybrid retriever combines dense and sparse rankings using Reciprocal Rank Fusion:

$$\text{score}(d) = \sum_{r \in R} \frac{1}{k + \text{rank}_r(d)}, \tag{1}$$

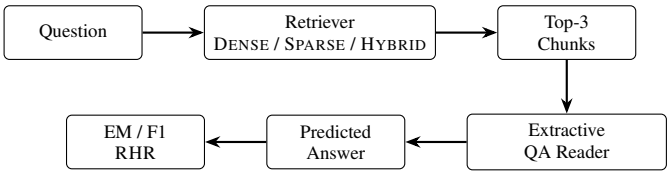


Figure 1: Pipeline architecture. The retriever is the only component changed across systems.

where $k = 60$ is the smoothing constant. RRF avoids directly mixing cosine similarities and BM25 scores, whose scales and distributions differ. Instead, it treats each retriever as a ranker and rewards documents that appear high in either or both lists.

Hybrid retrieval is expected to benefit from complementarity. When one retriever finds relevant chunks missed by the other, fusion can improve recall. However, if both retrievers already retrieve the same answer-bearing chunks, RRF adds little; if fusion changes evidence ordering, it can slightly reduce downstream F1.

3 System Architecture

The same data loader, chunker, answer generator, and evaluator are used for every retrieval method. Only the retrieval module changes across the three pipelines.

3.1 Data Loading

The project supports SQuAD and TriviaQA through a unified schema: each example contains a question, a context passage, and a list of gold answer strings. SQuAD is passage-grounded; questions are written from a known paragraph and answers are verbatim spans. TriviaQA is trivia-like and includes aliases for valid answer strings, such as "Sinclair Lewis" and "Harry Sinclair Lewis". The experiment uses 600 corpus contexts and 300 evaluation queries per dataset.

3.2 Chunking

Contexts are split into overlapping token windows of 250 tokens with 50-token overlap, using the dense encoder's tokenizer. Overlap reduces the probability that an answer span is split across chunk boundaries. Trailing fragments shorter than 10 tokens are discarded to avoid uninformative retrieval slots.

3.3 Retrievers

The dense retriever encodes all chunks into L2-normalised embeddings, builds a FAISS flat inner-product index, and returns the top-k chunks by cosine similarity. The sparse retriever tokenizes text using a lowercase word regex and ranks chunks with BM25 Okapi. The hybrid retriever over-fetches $\min(3k, 50)$ candidates from each retriever and applies RRF before returning the final top-k list.

Dataset	Style	Corpus	Queries
SQuAD	Lexical passage QA	600	300
TriviaQA	Paraphrased trivia QA	600	300

Table 1: Datasets and experimental scale. Expected winners are BM25 for SQuAD and dense retrieval for TriviaQA.

3.4 Answer Generator

The generator is `deepset/roberta-base-squad2`, an extractive QA model that predicts an answer span from retrieved text. Because it cannot generate text outside the retrieved chunks, failures can be attributed to either missing evidence or incorrect span selection. Each retrieved chunk is evaluated independently, and the span with the highest combined start and end logit score is returned.

4 Experimental Setup

4.1 Pipelines

Three pipelines are evaluated under identical conditions:

- DENSE: SentenceTransformer bi-encoder plus FAISS inner-product search.
- SPARSE: BM25 Okapi over tokenized chunks.
- HYBRID: Reciprocal Rank Fusion over dense and sparse ranked lists.

Each pipeline uses top-$k = 3$ retrieved chunks. The answer reader is held constant across all pipelines.

4.2 Evaluation Metrics

Exact Match (EM) measures whether the normalized prediction exactly matches any gold answer string after lowercasing and removing articles and punctuation. F1 gives partial credit based on token overlap and is more forgiving for span boundary variation. Retrieval Hit Rate (RHR) measures whether any retrieved chunk contains a normalized gold answer string, directly distinguishing retrieval failures from reader failures. Aggregate metrics use non-parametric bootstrap 95% confidence intervals with 1,000 resamples.

4.3 Corrected RHR Definition

The RHR implementation checks whether any gold answer string is a substring of any retrieved chunk, not merely whether the chunk overlaps with the gold context paragraph. This corrected definition makes RHR a true diagnostic: high RHR with low F1 implies the generator is the bottleneck; low RHR implies retrieval must be improved first.

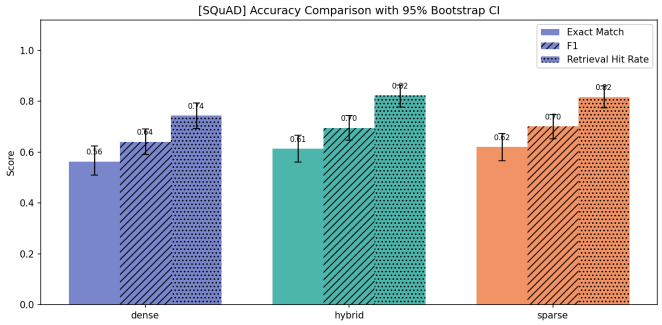


Figure 2: SQuAD accuracy comparison with 95% bootstrap confidence intervals.

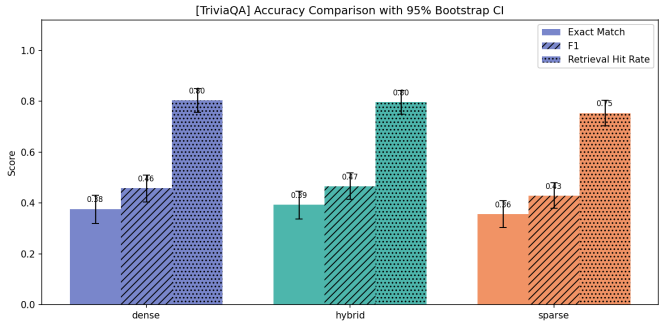


Figure 3: TriviaQA accuracy comparison with 95% bootstrap confidence intervals.

Pipe	EM	F1	F1 CI	RHR	Lat.
DENSE	0.563	0.641	.590–.691	0.743	**.712**
HYBRID	0.613	0.695	.648–.743	**0.823**	.734
SPARSE	**0.620**	**0.701**	.652–.747	0.817	.758

Table 2: SQuAD results, $n = 300$. Sparse leads on EM/F1; hybrid leads on RHR.

Pipe	EM	F1	F1 CI	RHR	Lat.
DENSE	0.377	0.458	.404–.510	**0.803**	**1.343**
HYBRID	**0.393**	**0.465**	.415–.519	0.797	1.415
SPARSE	0.357	0.429	.379–.480	0.753	1.398

Table 3: TriviaQA results, $n = 300$. Hybrid leads on F1 and EM; dense leads on RHR.

5 Results

5.1 SQuAD Aggregate Accuracy

Figure 2 and Table 2 report aggregate SQuAD results. Sparse and hybrid retrieval achieve the highest mean F1 scores, both ahead of dense retrieval. All three pipelines have median F1 of 1.0, confirming that many SQuAD questions are answered perfectly regardless of retriever. Differences emerge mainly in the failure tail. Sparse and hybrid lead on RHR, while dense retrieval's lower RHR indicates that it retrieves fewer answer-containing chunks on lexically rich SQuAD questions.

5.2 TriviaQA Aggregate Accuracy

Figure 3 and Table 3 show that the ranking reverses on TriviaQA. Hybrid leads on F1 and EM, closely followed by dense, while sparse drops to the lowest F1. This confirms the expected hypothesis: BM25's keyword-matching advantage weakens on paraphrased trivia-style questions. Overall F1 is substantially lower than SQuAD for all pipelines, indicating that TriviaQA is harder for this extractive reader.

5.3 Per-Query F1 Distributions

Mean scores can conceal distributional differences. On SQuAD, all three pipelines share a median F1 of 1.0, but dense has a larger zero-F1 fraction (28.0%) than hybrid (22.7%) and sparse (23.0%). On TriviaQA, distributions are much more spread: median F1 is 0.367 for dense, 0.400 for hybrid, and 0.0 for sparse. Zero-F1 fractions are 48.0%, 47.0%, and 50.7%, respectively.

5.4 Failure Mode Analysis

The failure-mode breakdown separates three outcomes: retrieval succeeds and generation succeeds, retrieval succeeds but generation fails, and retrieval fails outright. On SQuAD, dense retrieval fully succeeds on 56.3% of queries versus 61–62% for hybrid and sparse. Dense also has the highest outright retrieval failure rate (25.7%).

On TriviaQA, all pipelines achieve only 36–39% fully correct outcomes. Critically, 40–43% of queries experience retrieval success but generation failure: the answer was retrieved, but the extractive model could not produce a correct span. Sparse retrieval has the highest outright retrieval failure rate (24.7%), reflecting vocabulary mismatch on paraphrased questions.

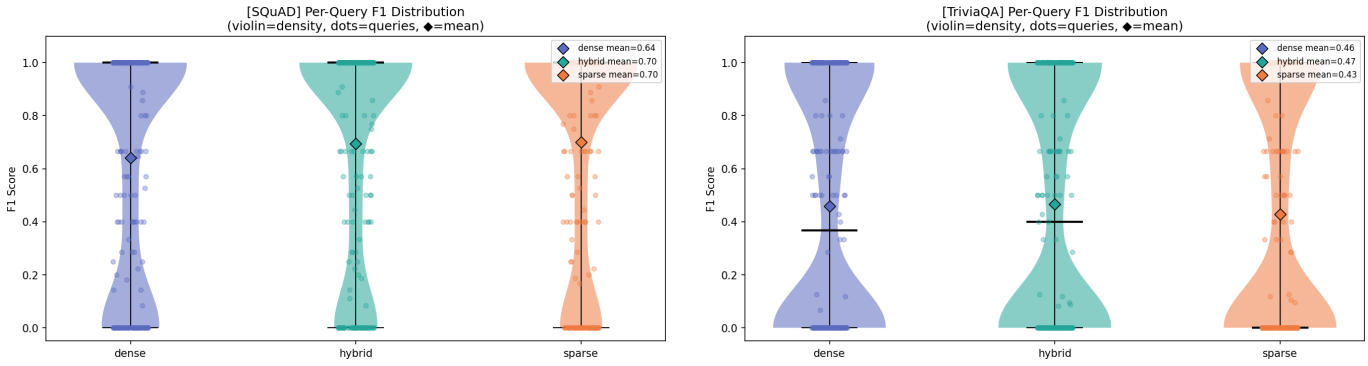
5.5 Latency Analysis

Table 4 reports average latency per query, averaged across both datasets. Generation dominates total runtime, accounting for most per-query time. Retrieval latency differences are small in absolute terms: all pipelines differ by about 50ms in total latency at this corpus scale.

6 Analysis

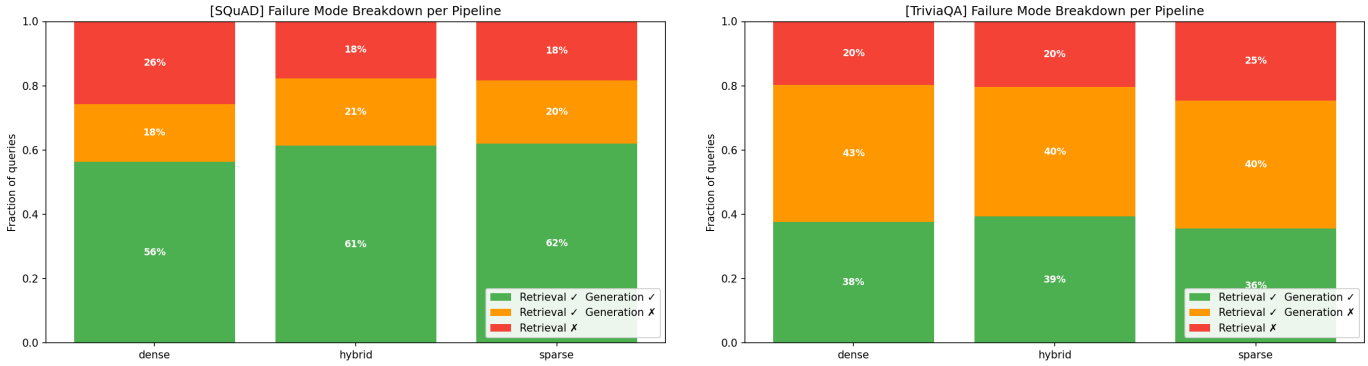
6.1 Cross-Dataset Comparison

Figure 7 shows the generalisation gap, TriviaQA F1 minus SQuAD F1. All gaps are negative, confirming that paraphrased trivia-style QA is harder. Dense has the smallest gap (-0.183), hybrid is intermediate (-0.230), and sparse has the largest gap (-0.272). This pattern confirms the core hypothesis: sparse retrieval is preferred for lexically overlapping QA but does not generalise as well to paraphrased questions. Dense retrieval is



(a) SQuAD **(b)** TriviaQA

Figure 4: Per-query F1 distributions. TriviaQA has a much heavier zero-F1 tail than SQuAD.



(a) SQuAD **(b)** TriviaQA

Figure 5: Failure mode breakdown. TriviaQA is dominated by generator failures after successful retrieval.

Pipeline	Retrieval	Generation	Total
DENSE	**0.014s**	1.014s	**1.028s**
HYBRID	0.056s	1.018s	1.074s
SPARSE	0.039s	1.039s	1.078s

Table 4: Average per-query latency across SQuAD and TriviaQA. Generation dominates runtime.

more robust across regimes, while hybrid is strongest when the query distribution is mixed.

6.2 Best Retriever by Criterion

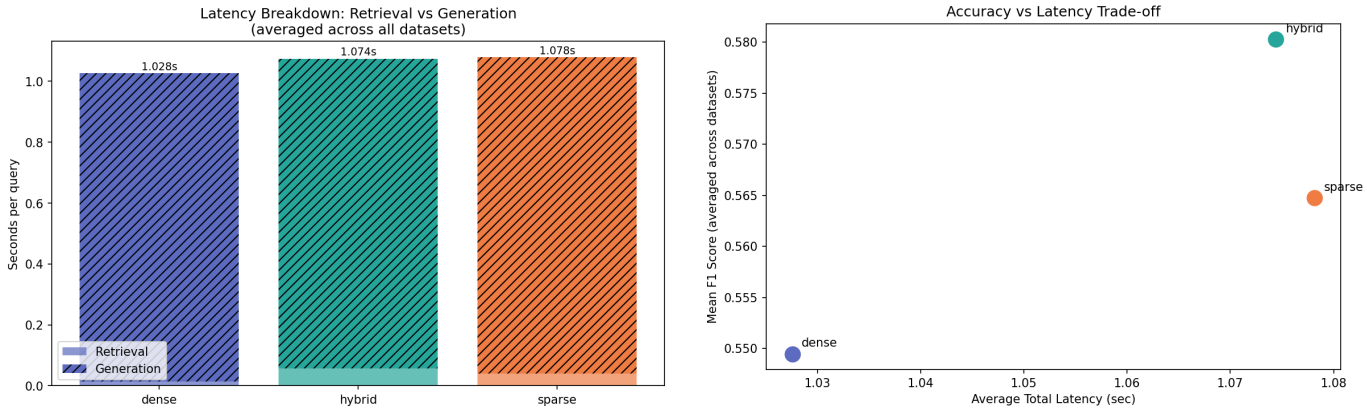
For SQuAD accuracy, sparse and hybrid are effectively tied, with BM25's lexical matching advantage driving the result. For TriviaQA accuracy, hybrid leads narrowly over dense, and dense outperforms sparse by about three F1 points. For generalisation, dense is the most robust individual retriever because it has the smallest cross-dataset F1 gap. For overall practical use, hybrid is strongest: it has the highest mean F1 across datasets (0.580), strong RHR values, and only a small latency difference relative to individual retrievers.

6.3 Generator as the Primary Bottleneck

Across both datasets, RHR substantially exceeds end-to-end F1. On TriviaQA, dense retrieval reaches RHR=0.803 but only F1=0.458, a gap that cannot be explained by retrieval failure alone. The failure-mode plots confirm this directly: 40–43% of TriviaQA queries have correct retrieval but failed generation. This implies that upgrading the generator, using a model trained on TriviaQA-style answers, or allowing cross-chunk evidence aggregation would likely yield larger gains than further retrieval tuning alone.

6.4 Per-Query Pipeline Disagreement

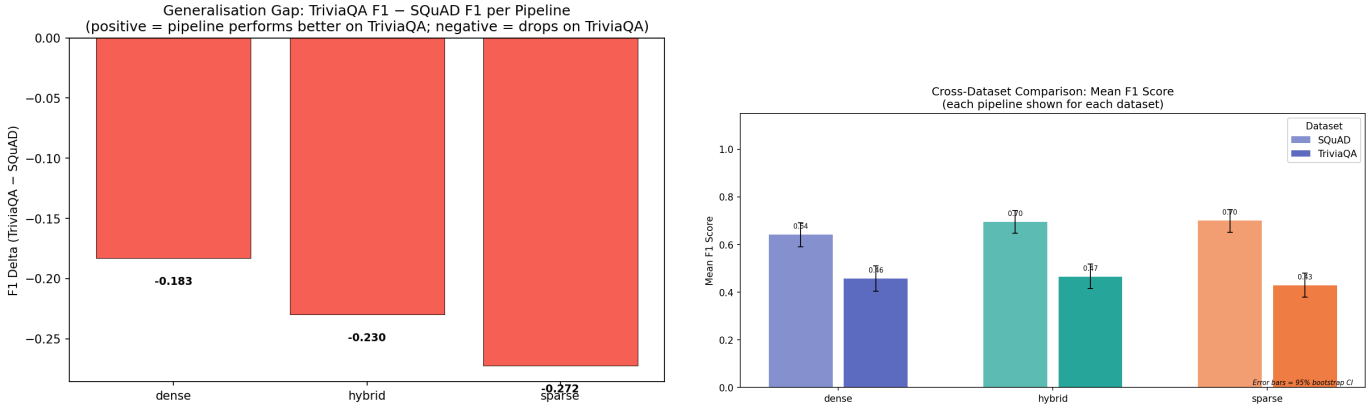
Dense and sparse retrieval disagree on a meaningful minority of queries. On SQuAD, sparse beats dense on 44/300 queries, dense beats sparse on 24/300, and 232 are tied; 49 queries fail under both. On TriviaQA, dense beats sparse on 44/300, sparse beats dense on 32/300, and 224 are tied; 115 fail under both. The higher mutual failure rate on TriviaQA shows that neither retriever fully covers the distribution and further motivates hybrid retrieval and generator improvement.



(a) Latency breakdown

(b) Accuracy-latency trade-off

Figure 6: Latency and accuracy trade-off. Hybrid gives the best mean F1, while dense is slightly fastest overall.



(a) Generalisation gap

(b) Cross-dataset F1

Figure 7: Cross-dataset analysis. Sparse retrieval suffers the largest F1 drop on TriviaQA.

7 Limitations

Generator bias. The generator, `deepset/roberta-base-squad2`, is fine-tuned on SQuAD-style extractive spans. On TriviaQA, aliases and varied surface forms can cause EM/F1 to underestimate retrieval quality even when the retrieved passage is semantically useful.

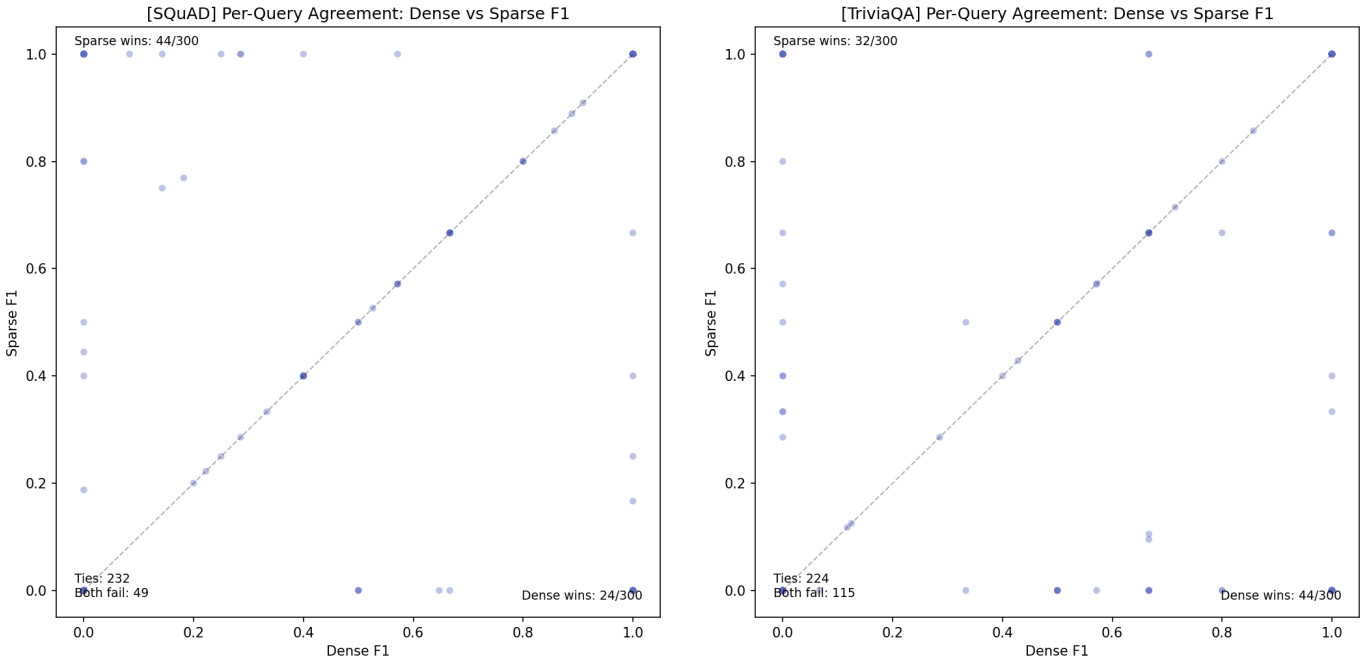
Fixed top-k and chunk size. All results use top-$k = 3$ and 250-token chunks with 50-token overlap. Retrieval quality is sensitive to both parameters. The codebase includes top-k and chunk-size ablations, but the primary analysis fixes them to keep the comparison interpretable.

Corpus scale. Each dataset uses 600 corpus contexts, which is small compared with production retrieval settings. At larger scale, dense approximate nearest-neighbor search and BM25 may show different accuracy-latency trade-offs.

RRF hyperparameter. The hybrid method uses the standard RRF smoothing constant $k = 60$. Tuning k, or replacing RRF with a learned reranker, could change the hybrid ranking.

8 Conclusion

This report presents a two-dataset benchmark comparing dense, sparse, and hybrid retrieval in RAG. The results support four conclusions. First, dataset regime determines the best retriever: sparse BM25 is best on SQuAD-style lexical QA, while dense and hybrid retrieval are stronger on TriviaQA-style semantic QA. Second, hybrid is the safest practical choice, achieving the highest mean F1 across both datasets and the best TriviaQA F1. Third, sparse retrieval does not generalise as well, suffering the largest cross-dataset F1 drop. Fourth, the generator is the primary bottleneck on TriviaQA: many queries retrieve answer-bearing evidence but still fail answer extraction. Future work should therefore prioritize stronger generators, cross-chunk answer aggregation, and retrieval hyperparameter ablations.



(a) SQuAD

(b) TriviaQA

Figure 8: Dense versus sparse per-query F1 disagreement. TriviaQA has many more mutual failures.

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